Input system:
Dependent variables:$p_1, g_1, g_2, p_2, p_3, g_3$
Independent variables:t

$$\frac{dg_1}{dt} = p_1$$
$$\frac{dg_2}{dt} = p_2$$
$$\frac{dg_3}{dt} = p_3$$
$$\frac{dp_1}{dt} = (-1)k^2(m_0 + m_1)g_1\text{isqrt}(g_1^2 + g_2^2 + g_3^2)^3$$
$$\frac{dp_2}{dt} = (-1)k^2(m_0 + m_1)g_2\text{isqrt}(g_1^2 + g_2^2 + g_3^2)^3$$
$$\frac{dp_3}{dt} = (-1)k^2(m_0 + m_1)g_3\text{isqrt}(g_1^2 + g_2^2 + g_3^2)^3$$

==========

Result:
Dependent variables:$p_1, p_3, g_1, g_2, g_3, p_2, s_1$
Independent variables:t
Additional variables:

$$s_1 = \text{isqrt}(g_1^2 + g_2^2 + g_3^2)$$
$$\frac{dg_1}{dt} = p_1$$
$$\frac{dg_2}{dt} = p_2$$
$$\frac{dg_3}{dt} = p_3$$
$$\frac{dp_1}{dt} = (-1)s_1^3g_1k^2m_0 + (-1)s_1^3g_1k^2m_1$$
$$\frac{dp_2}{dt} = (-1)s_1^3g_2k^2m_0 + (-1)s_1^3g_2k^2m_1$$
$$\frac{dp_3}{dt} = (-1)s_1^3g_3k^2m_0 + (-1)s_1^3g_3k^2m_1$$
$$\frac{ds_1}{dt} = (-1)s_1^3g_1p_1 + (-1)s_1^3g_2p_2 + (-1)s_1^3g_3p_3$$